

# Zach Landry

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Boston, Massachusetts

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## Education

Northeastern University, Boston, MA

September 2024 - April 2028

Candidate for Bachelor of Science in Electrical Engineering and Physics

GPA 4.0 | Dean's List

**Relevant Coursework:** Fundamentals of Electronics, Embedded Design, Computing Fundamentals, Calculus III, Differential Equations & Linear Algebra, Quantum Mechanics, Electricity & Magnetism 1

**Current Coursework:** Electronic Design, Linear Systems, Classical Dynamics, Advanced Quantum Mechanics

**Activities:** IEEE, Generate, College of Engineering Tutoring, Downhillers, NUHOC

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## Skills

**Electronics:** Breadboarding, Soldering (THT/SMD), Analog Circuit Design, Embedded Hardware Design, PCB layout, DMM, Oscilloscope, Function Generator

**Software:** KiCad, LTspice, AutoCAD, SolidWorks, C++, Python, Arduino, MATLAB, Excel

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## Work Experience

Electrical Engineering Co-op | Massa | Hingham, Massachusetts

January 2026 - June 2026

- Designed and built an ultra-low-power PIC-based embedded system on a custom PCB, integrating an external custom designed sensor module for field data acquisition
- Developed a multi-stage analog filter chain to condition sensor signals for high-accuracy ADC sampling
- Diagnosed and resolved complex PCB failures using oscilloscopes, signal tracing, and component-level debugging

Engineering Research Assistant | Northeastern University | Boston, Massachusetts

June 2025 - Present

- Built an open source AI integrated pair of glasses to be researched
  - Design a waveguide-based lens display system and developed embedded graphics code for on-lens visualization
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## Personal Projects

Analog Synthesizer

May 2026 - Present

- Designed a triangle-core VCO feeding a state-variable VCF and an OTA-based VCA
- Built an ADSR envelope generator and exponential converter for control-voltage shaping
- Designed an analog AGC stage for consistent output amplitude across frequencies
- Implemented all analog blocks using discrete components and op-amps/OTAs

Hexapod

May 2025 - July 2025

- Designed and built a 3D-printed hexapod robot
- Wired 18 servo motors to an ESP-32 microcontroller with Adafruit drivers
- Coded a gait algorithm for multi-directional movement in the Arduino IDE (written in C++)

Differential Equation Visualizer

February 2025 - April 2025

- Built an interactive frontend for visualizing ODE's
- Implemented a backend solver which supports arbitrary equation inputs
- Included slope fields for first-order equations and special step and impulse functions